

PRATIM BAIDYA

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EXPERIENCE

IoTALL Labs Private Limited / Kolkata / Jan – Jun 2025

- Worked on building and optimizing computer vision models for gesture and posture detection. Used libraries like OpenCV and MediaPipe, YOLO, Keras, Sk-Learn, Matplotlib, Seaborn. Applied callbacks, early stopping and data augmentation techniques to reduce overfitting. Achieved closed to 90% accuracy.
- Developed an audio classification model for detecting and classifying different sounds inside a house using pretrained YAMNet model. Used libraries like Keras, Librosa, Sk-Learn, Matplotlib, Seaborn.
- Conducted product and market analysis for IoT products and reporting.
- Built automated Post and Messaging for different social media platforms using Python Selenium library for digital marketing.
- Explored small vision language models for image generation and video description generation for digital marketing.

EDUCATION

- BS Degree Program | IIT Madras | Data Science | Present CGPA – 8.34 | 2023 – Present [\[Progress\]](#)
- B. Tech | MAKAUT | Mechanical Engineering | CGPA – 9.05 | 2019 – 2022 [\[Certificate\]](#)

PROJECTS

- Heart Disease Classification. [\[Link\]](#)
 - Libraries used – Sk-Learn, NumPy, Pandas, Matplotlib, Seaborn
 - Analysis of different feature to look for trends and patterns among them.
 - ML Models – Logistic Regression, K-NN, Random Forest (with tuned hyper-parameters)
 - Model evaluation with Classification report, Confusion Matrix, ROC-AUC Curve.
- Brain Cancer Classification. [\[Link\]](#)
 - Libraries used – PyTorch, Sk-Learn, NumPy, Pandas, Matplotlib.
 - Trained a custom CNN model and Used Resnet50 pre-trained model.
 - Achieved 94% accuracy with almost no overfitting on a dataset with balanced classes.
 - Used callback methods, early stopping and data augmentation techniques to reduce over fitting.
- Sarcasm Detection from News Headlines. [\[Link\]](#)
 - Libraries used – Pytorch, Sk-Learn, Matplotlib
 - Used one of the pretrained BERT model and its tokenizer.

SKILLS

- Python – OOPs, NumPy, Pandas, Matplotlib, Scikit-Learn, PyTorch, Seaborn, OpenCV, Hugging Face
- Tools - Excel, Power BI, SQL

AREA OF STUDY

- Data Science - Data Cleaning and Feature Engineering, Data Analysis and Visualization, Machine Learning (Regression - Linear, Ridge, Lasso, Kernel, Polynomial; Classification - Logistic, KNN, Random Forest; Unsupervised - PCA, K-Means), Model Tuning, Model Testing and Evaluation
- Statistics – Probability, Descriptive Statistics, Inferential Statistics
- Linear Algebra

COURSES

- WorldQuant University – Applied Data Science Lab [\[Credentials\]](#)

LANGUAGE

- English, Hindi, Bengali